Capstone controlled experiments

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

THE CONNECTED WEEK

One complete controlled-experiment record.

Keep fixed settings, changed rate, all checkpoints and validation choice together.

Keep one habit: Recall → W1–W3/W5 → independent comparison → project history → delayed return.

MAIN LESSON · SECTION 1

Recall · Retrieve before notes

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Capstone hypothesis and baseline. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Efficiency and resource measurement. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Transfer learning. Give an example and a mistake that the idea helps you avoid.

- Use this reference for the original W1–W3 and W5 exercises. The separately named independent case supplies its own changed facts; use those facts for that attempt before feedback.

- Continue with **training** (1,2) and (2,4), validation (3,6) and (4,8), and **prediction** max(0, a*x). Compare learning rates 0.1 and 0.2. Each run starts at a = 1, has no bias and takes two full-batch updates.

  Keep the loss, data order and arithmetic fixed. This toy model has no randomness; repeating the exact run checks replay, not variation across seeds.

- Record the comparison as an explicit revision to the one-candidate Week 33 plan before viewing new results. Validation may guide selection; sealed final data must not.

  Select lower validation MSE, with a tie going to learning rate 0.1.

Capstone hypothesis and baseline

> **My remembered idea, example and correction**

Efficiency and resource measurement

> **My remembered idea, example and correction**

Transfer learning

> **My remembered idea, example and correction**
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Lesson reference for workbook practice

MAIN LESSON · SECTION 5

W1 · Explain the current example

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Main versioned comparison: prediction ReLU(a*x), no bias, initial a1, two full-batch MSE updates; training (1, 2), (2, 4), validation (3, 6), (4, 8). Active-region gradient 5(a–2). Rate .1: gradient–5, weight 1.5, then gradient–2.5, weight 1.75; final train MSE .15625, validation predictions 5.25/7, squared errors .5625/1, MSE .78125. Rate .2: gradient–5, weight 2, then gradient 0, weight 2; train/validation MSE 0.

  Baseline a1 train MSE 2.5, validation 12.5. Lower validation MSE selects .2; ties go to .1. Only two distinct validation cases. Separate supported .3 path 1→2.5→1.75 has same final score but different intermediate state and is not retroactively a selection candidate. Separate supplied picture flags A [1, 1, 0, 1, 0, 1], B [1, 1, 1, 1, 0, 1]: correct 4/6 versus 5/6; case 3 differs, case 5 fails both.

  Flags are not predicted labels; no actual image-model run is supplied. Final cases remain outside selection; no timings or seed variation measured.

- List six settings held fixed and the single intended changed factor.

My reasoning and evidence

MAIN LESSON · SECTION 6

W2 · Trace the calculation

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Main versioned comparison: prediction ReLU(a*x), no bias, initial a1, two full-batch MSE updates; training (1, 2), (2, 4), validation (3, 6), (4, 8). Active-region gradient 5(a–2). Rate .1: gradient–5, weight 1.5, then gradient–2.5, weight 1.75; final train MSE .15625, validation predictions 5.25/7, squared errors .5625/1, MSE .78125. Rate .2: gradient–5, weight 2, then gradient 0, weight 2; train/validation MSE 0.

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  Flags are not predicted labels; no actual image-model run is supplied. Final cases remain outside selection; no timings or seed variation measured.

- Compute both updates for rates 0.1 and 0.2 using gradient 5*(a-2).

My reasoning and evidence

MAIN LESSON · SECTION 7

W3 · Repair a misconception

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Main versioned comparison: prediction ReLU(a*x), no bias, initial a1, two full-batch MSE updates; training (1, 2), (2, 4), validation (3, 6), (4, 8). Active-region gradient 5(a–2). Rate .1: gradient–5, weight 1.5, then gradient–2.5, weight 1.75; final train MSE .15625, validation predictions 5.25/7, squared errors .5625/1, MSE .78125. Rate .2: gradient–5, weight 2, then gradient 0, weight 2; train/validation MSE 0.

  Baseline a1 train MSE 2.5, validation 12.5. Lower validation MSE selects .2; ties go to .1. Only two distinct validation cases. Separate supported .3 path 1→2.5→1.75 has same final score but different intermediate state and is not retroactively a selection candidate. Separate supplied picture flags A [1, 1, 0, 1, 0, 1], B [1, 1, 1, 1, 0, 1]: correct 4/6 versus 5/6; case 3 differs, case 5 fails both.

  Flags are not predicted labels; no actual image-model run is supplied. Final cases remain outside selection; no timings or seed variation measured.

- Recompute validation MSE for the final weights and apply the written **selection rule**. How many distinct validation rows were used?

My reasoning and evidence

MAIN LESSON · SECTION 9

W5 · Explain the limitation

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Main versioned comparison: prediction ReLU(a*x), no bias, initial a1, two full-batch MSE updates; training (1, 2), (2, 4), validation (3, 6), (4, 8). Active-region gradient 5(a–2). Rate .1: gradient–5, weight 1.5, then gradient–2.5, weight 1.75; final train MSE .15625, validation predictions 5.25/7, squared errors .5625/1, MSE .78125. Rate .2: gradient–5, weight 2, then gradient 0, weight 2; train/validation MSE 0.

  Baseline a1 train MSE 2.5, validation 12.5. Lower validation MSE selects .2; ties go to .1. Only two distinct validation cases. Separate supported .3 path 1→2.5→1.75 has same final score but different intermediate state and is not retroactively a selection candidate. Separate supplied picture flags A [1, 1, 0, 1, 0, 1], B [1, 1, 1, 1, 0, 1]: correct 4/6 versus 5/6; case 3 differs, case 5 fails both.

  Flags are not predicted labels; no actual image-model run is supplied. Final cases remain outside selection; no timings or seed variation measured.

- Run A uses rate 0.1 for two updates; run B uses rate 0.2 for six updates and different **training** rows. Can the difference be attributed to rate alone? Propose a repair.

My reasoning and evidence

MAIN LESSON · SECTION 4

Try alone · Selected transfer

> Keep this task's inputs,rule and response together.

Complete task and evidence

- For **training** (1,3),(2,6) and validation (3,9),(4,12), start a = 1. Compare rates 0.1 and 0.2 for two updates. Gradient is 5*(a-3). Compute final weights and validation MSE, then select lower MSE; ties go to 0.1.

- Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Independent case: Run a new comparison independently

My independent explanation, trace and limits

MAIN LESSON · SECTION 10

Project · Keep the mathematical evidence

> Keep this task's inputs,rule and response together.

Complete task and evidence

- For Image-pattern Network Investigation, use this week's independent task as a rehearsal for the final **evidence** pack: For **training** (1,3),(2,6) and validation (3,9),(4,12), start a = 1.

  Compare rates 0.1 and 0.2 for two updates. Gradient is 5*(a-3). Compute final weights and validation MSE, then select lower MSE; ties go to 0.1.

  Keep actual **observation**s separate from expected results, preserve failures, compare a baseline and explain the limits.

> **My project change, supporting evidence and remaining limit**

MAIN LESSON · SECTION 11

Feedback · Repair and return

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use W2: Trace both runs for Capstone controlled experiments. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

> **The step I completed without help, and the hint I needed**

▼ Extra practice · Individual reasoning check

REFERENCE · SECTION 3

Extra practice · Individual reasoning check

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use the worked case for Capstone controlled experiments. Proposed explanation: "More runs mean more validation examples". Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Earlier W4 response copy

REFERENCE · SECTION 8

Reference · Earlier W4 response copy

> Keep this task's inputs,rule and response together.

Complete task and evidence

- For **training** (1,3),(2,6) and validation (3,9),(4,12), start a = 1. Compare rates 0.1 and 0.2 for two updates. Gradient is 5*(a-3). Compute final weights and validation MSE, then select lower MSE; ties go to 0.1.

> **My reasoning and evidence**

▼ **Extra practice · Pictured case**

REFERENCE · SECTION 12

Extra practice · Pictured case

> Keep this task's inputs,rule and response together.

Complete task and evidence

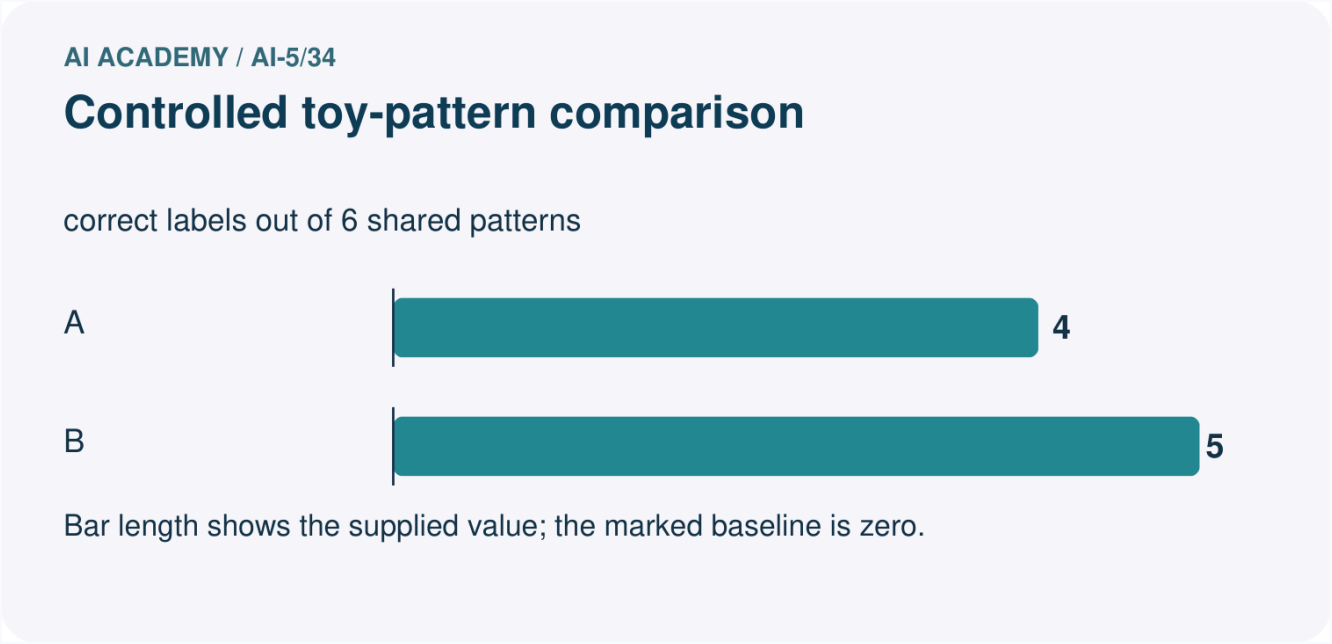
- One controlled network change. The capstone compares two fixed networks on the same six labelled toy patterns. Only one feature connection is added.

- A scores [1,1,0,1,0,1]; B scores [1,1,1,1,0,1]. Settings, labels and scoring remain fixed. Binary entries mean 1=correct and 0=incorrect.

- Compute the two correct counts and identify the changed case. Why would changing B's **training** data weaken the conclusion about its added connection?

- Support used: alone / hint / worked example. This record is practice **evidence**.
- Project connection: For the Image-pattern Network Investigation, save the labelled trace and the changed-case explanation beside the experiment record.

  State exactly what this example supports before linking it to the wider capstone claim.



Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-5/34

Capstone controlled experiments

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Identify the intervention

B adds the centre-feature connection; all other declared settings stay fixed.

2. Count outcomes

A is 4/6 correct; B is 5/6.

3. Inspect paired cases

Only case three changes correctness; case five fails for both.

CHANGE ONE THING

Change B's training set as well as the feature connection.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

My picture-based explanation and changed-case prediction

▼ Reference · Project milestone

REFERENCE · SECTION 13

Reference · Project milestone

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Project: Image-pattern Network Investigation
- Current milestone: Present and defend. Demonstrate the system, explain one failure and justify the next improvement with **evidence**.
- For the Image-pattern Network Investigation, save the labelled trace and the changed-case explanation beside the experiment record. State exactly what this example supports before linking it to the wider capstone claim.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ Extra practice · Predict,check and repair

REFERENCE · SECTION 14

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- One controlled network change. The capstone compares two fixed networks on the same six labelled toy patterns. Only one feature connection is added.
- Which labelled step matches this detail? Only case three changes correctness; case five fails for both.
- 1. Inspect paired cases
- 2. Identify the intervention
- 3. Count outcomes
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: Inspect paired cases Only case three changes correctness; case five fails for both.

Controlled experiments vary the specified component while retaining comparable cases. Pairing the outcomes by pattern shows where the score difference comes from instead of hiding it in an overall average.

Changed-case question and optional hint

- Changed-case practice: Change B's **training** set as well as the feature connection. Which conclusion is supported by the supplied facts?
- 1. This one example guarantees correct results for every future **input**.
- 2. We can decide the result without examining the changed **input** or the procedure.
- 3. The experiment no longer isolates the connection change.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: The experiment no longer isolates the connection change. A **training**-data difference supplies an alternative explanation for the observed result.

Repair and retain the record

Review Capstone hypothesis and baseline (ai-5/33). Goal: Write a testable capstone hypothesis and a credible baseline. Short practice: Name the intended change, the baseline and the measured outcome.

Why must both versions avoid **training** on the eight reserved labels?. Return to this week and explain what you changed.

Review Efficiency and resource measurement (ai-5/31). Goal: Calculate throughput and amortised time per example from repeated timings. Short practice: Why is 10 ms/case not necessarily a single-request latency?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Transfer learning (ai-5/28). Goal: Trace a reused feature extractor and a newly fitted **output** head. Short practice: Which part was trained for the new task?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

▼ **Reference · Six-picture response sheets**

REFERENCE · SECTION 15

Reference · Six-picture response sheets

> Choose the picture that helps explain the current step.

01. Capstone controlled experiments: the case

AI ACADEMY / AI-5/34 · 01 · LOOK AT THE FACTS

Capstone controlled experiments: the case

The situation

One controlled network change. The capstone compares two fixed networks on the same six labelled toy patterns. Only one feature connection is added.

Supplied evidence

A scores [1,1,0,1,0,1]; B scores [1,1,1,1,0,1]. Settings, labels and scoring remain fixed. Binary entries mean 1=correct and 0=incorrect.

Explain capstone controlled experiments with a traceable toy-network case and a bounded conclusion.

Follow the labelled picture with the matching explanation. Open the vector diagram

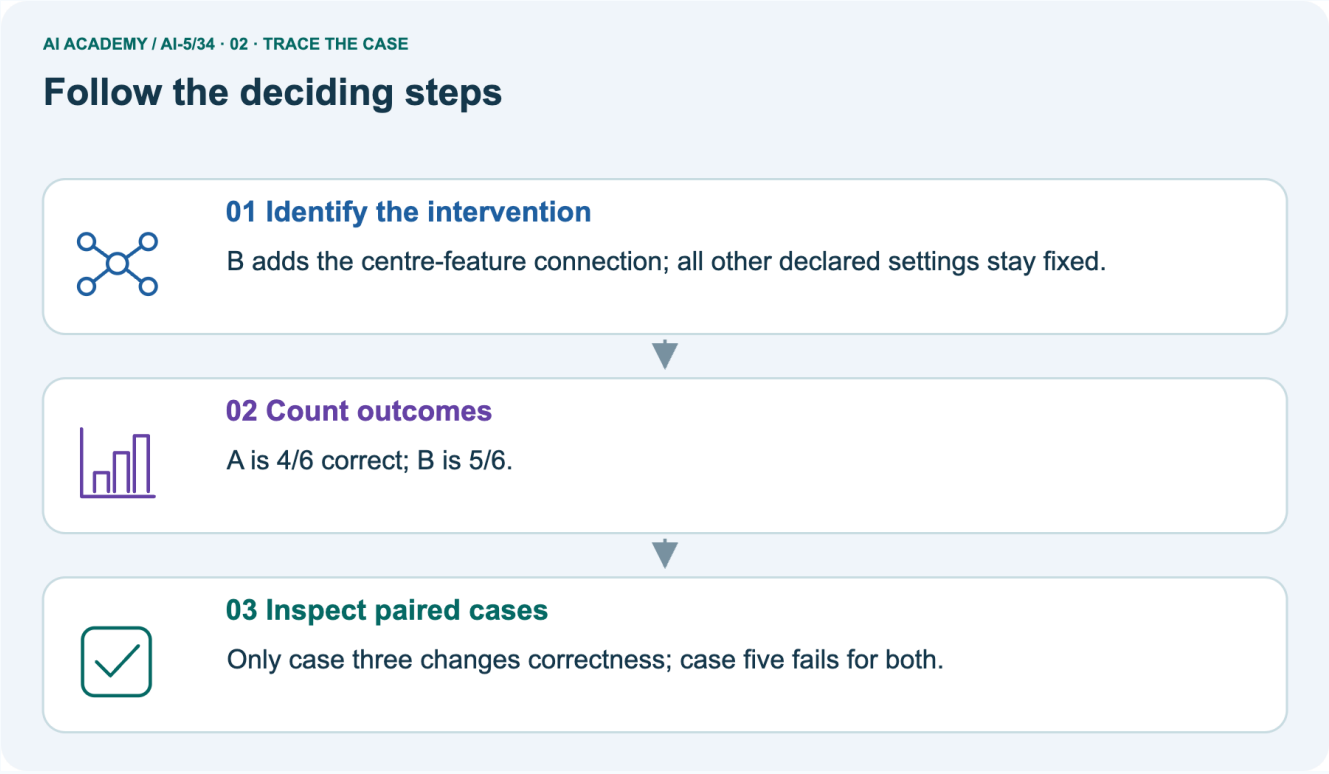
One controlled network change. The capstone compares two fixed networks on the same six labelled toy patterns. Only one feature connection is added. Supplied facts: A scores [1,1,0,1,0,1]; B scores [1,1,1,1,0,1].

Settings, labels and scoring remain fixed. Binary entries mean 1=correct and 0=incorrect.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



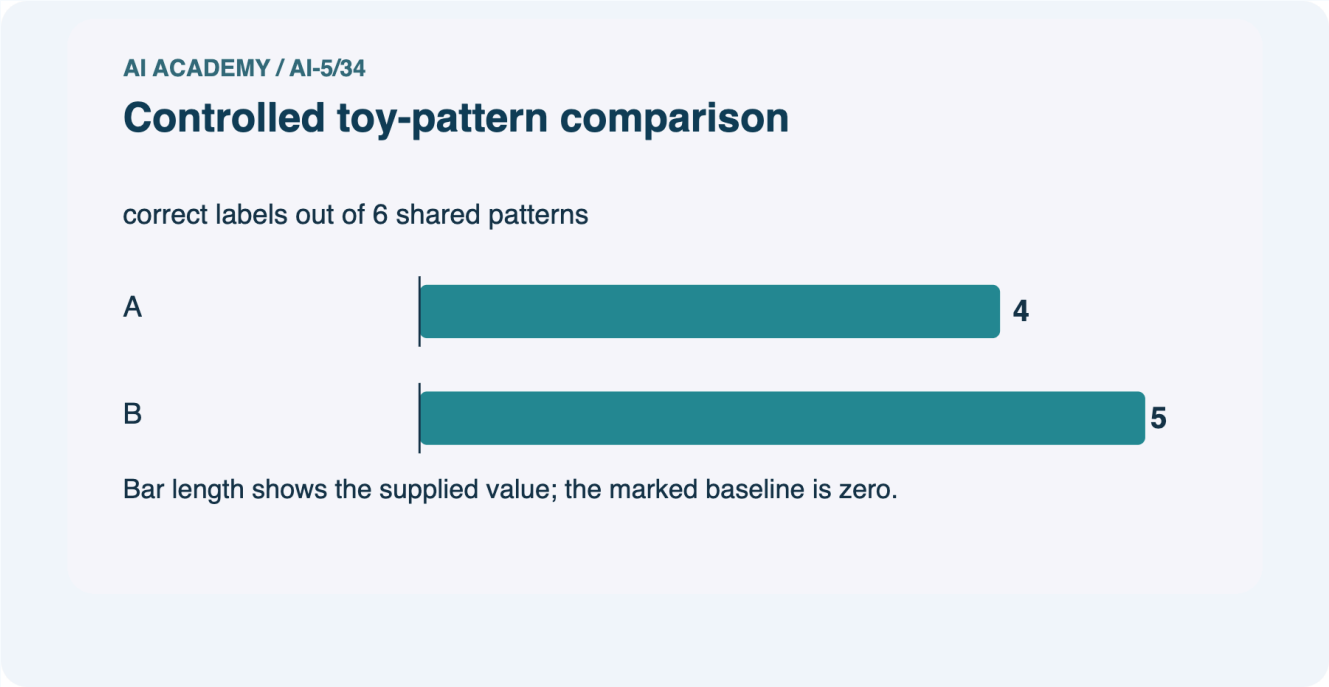
Follow the labelled picture with the matching explanation. Open the vector diagram

Identify the intervention: B adds the centre-feature connection; all other declared settings stay fixed. → Count outcomes: A is 4/6 correct; B is 5/6. → Inspect paired cases: Only case three changes correctness; case five fails for both.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Controlled toy-pattern comparison



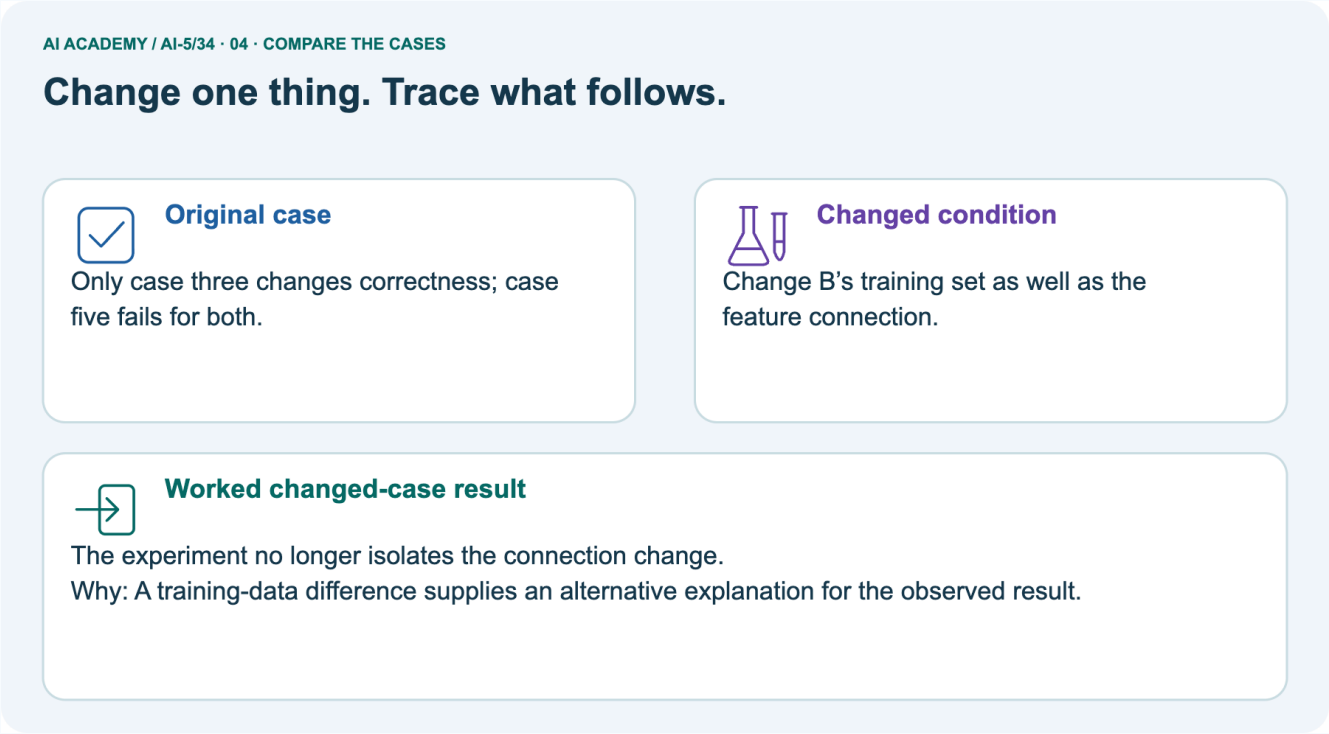
Follow the labelled picture with the matching explanation. Open the vector diagram

Controlled toy-pattern comparison. Units: correct labels out of 6 shared patterns. A: 4. B: 5.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

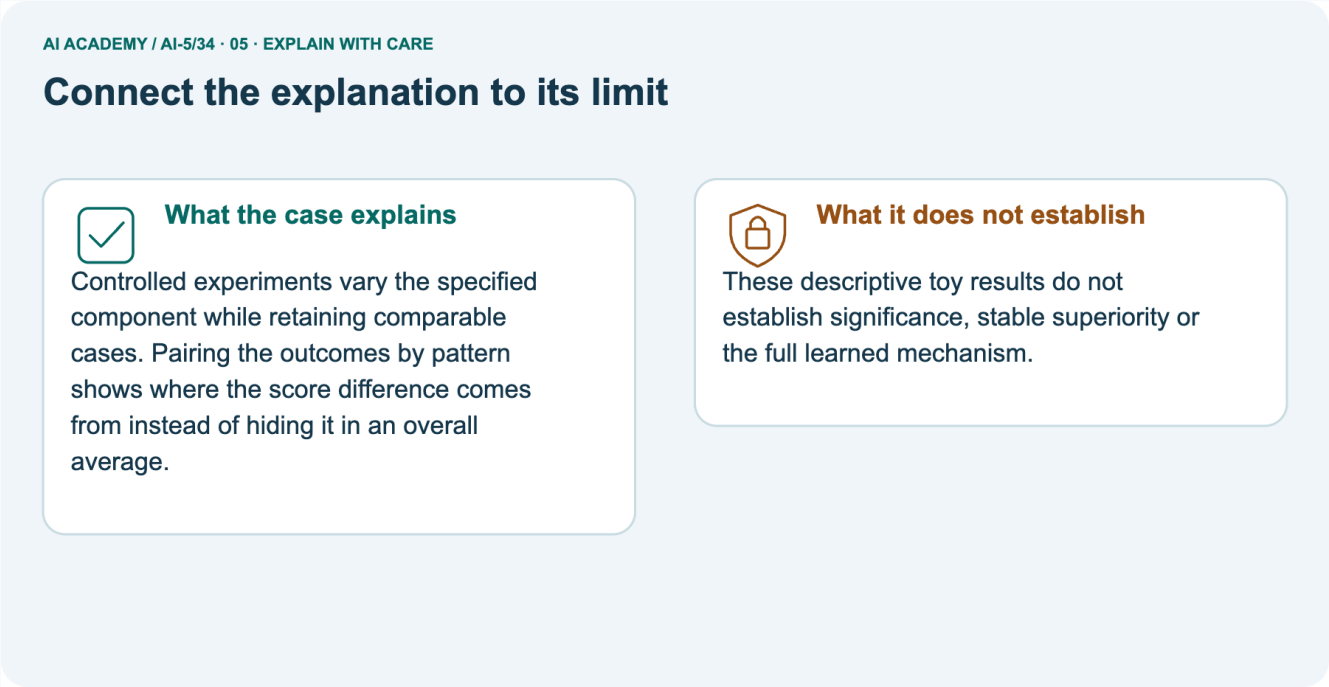
Original case: Only case three changes correctness; case five fails for both. Changed condition: Change B's **training** set as well as the feature connection.

Expected result: The experiment no longer isolates the connection change. Reason: A **training**-data difference supplies an alternative explanation for the observed result.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

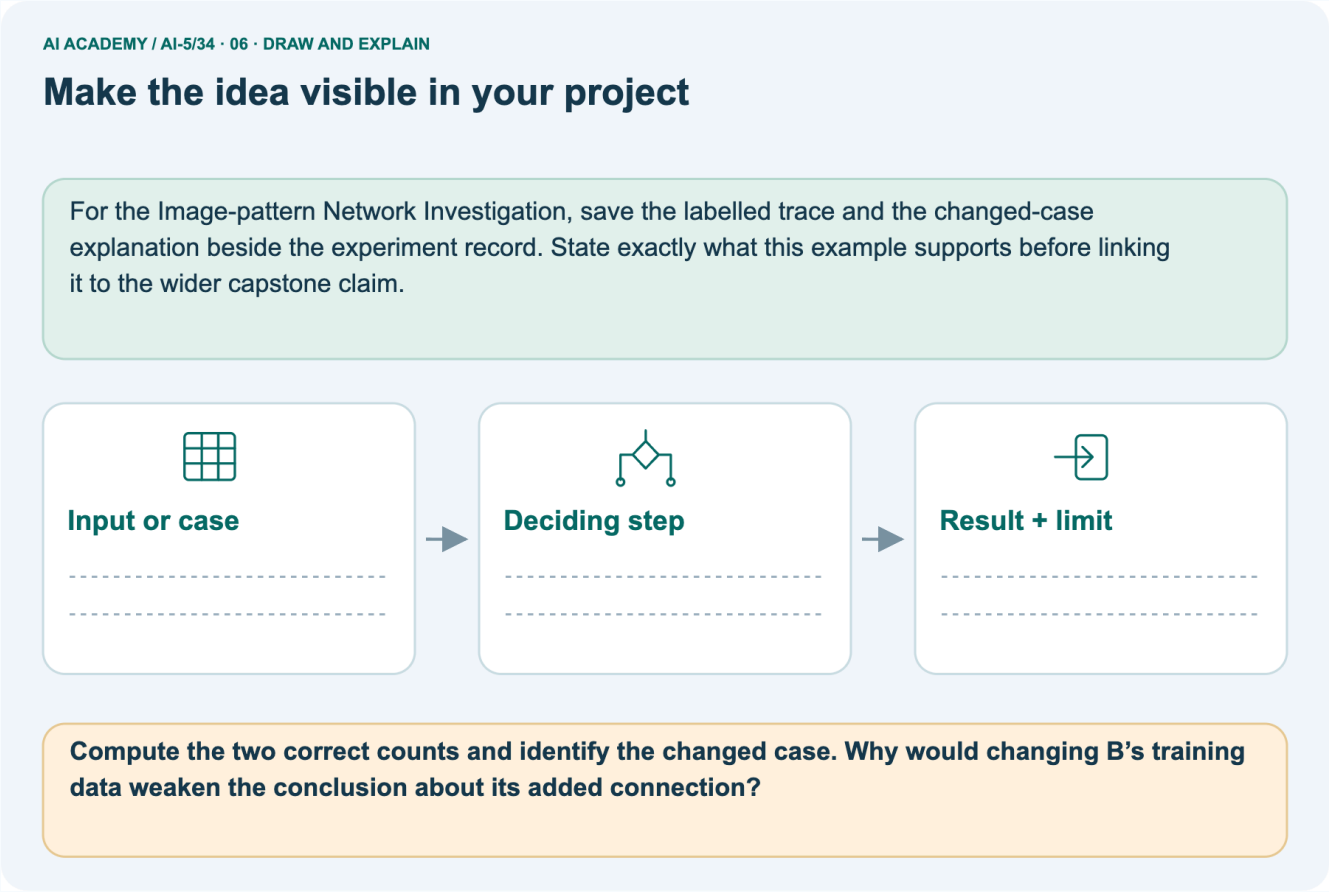
Controlled experiments vary the specified component while retaining comparable cases. Pairing the outcomes by pattern shows where the score difference comes from instead of hiding it in an overall average.

Limit: These descriptive toy results do not establish significance, stable superiority or the full learned **mechanism**.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Image-pattern Network Investigation, save the labelled trace and the changed-case explanation beside the experiment record. State exactly what this example supports before linking it to the wider capstone claim.

Draw your **input**, deciding step, result and limit. Supported question: Compute the two correct counts and identify the changed case. Why would changing B's **training** data weaken the conclusion about its added connection?

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Capstone controlled experiments: the case

02. Follow the deciding steps

03. Controlled toy-pattern comparison

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.